

DATA TYPES

1)Create a console application to accept the side of a square and print the area of the square.

```
using System;

class Program
{
    static void Main()
    {
        Console.Write("Enter the side of the square: ");

        double side = Convert.ToDouble(Console.ReadLine());

        double area = side * side;

        Console.WriteLine("Area of the square = " + area);
    }
}
```

OUTPUT

Enter the side of the square: 7.5

Area of the square = 56.25

2)Create a console application that finds the number of digits and alphabets in a given string.

```
using System;

class Program
{
    static void Main()
    {
        Console.Write("Enter a string: ");

        string str = Console.ReadLine();

        int alphabets = 0, digits = 0;
```

```

        foreach (char ch in str)
        {
            if (char.IsLetter(ch))
                alphabets++;
            else if (char.IsDigit(ch))
                digits++;
        }

        Console.WriteLine("Number of Alphabets: " + alphabets);
        Console.WriteLine("Number of Digits: " + digits);
    }
}

```

OUTPUT

Enter a string: CSharp2025

Number of Alphabets: 6

Number of Digits: 4

3) Create a console application to increment each character by one and toggle its case.

using System;

```

class Program
{
    static void Main()
    {
        Console.Write("Enter a string: ");
        string str = Console.ReadLine();
        string result = "";
        foreach (char ch in str)
        {
            char newChar = (char)(ch + 1);

```

```

        if (char.IsLetter(newChar))
        {
            if (char.IsUpper(newChar))
                newChar = char.ToLower(newChar);
            else
                newChar = char.ToUpper(newChar);
        }
        result += newChar;
    }

    Console.WriteLine("Result: " + result);
}
}

```

OUTPUT

Enter a string: xyz9

Result: YZ{:

4) Create a console application to accept a string and perform the below string manipulation on the same:

- 1. Print the string in reverse order.**
- 2. Extract part of the string from 2nd position till the end of the string.**
- 3. Replace any given character by '\$' and print the new string.**
- 4. Copy the string to another string variable, modify the data in the 2nd string variable and print both the strings.**

```
using System;
```

```
class Program
```

```
{
```

```
    static void Main()
```

```
{
```

```
    Console.Write("Enter a string: ");
```

```

string str = Console.ReadLine(
char[] arr = str.ToCharArray();
Array.Reverse(arr);
Console.WriteLine("Reversed String: " + new string(arr));
if (str.Length > 1)
    Console.WriteLine("Substring from 2nd position: " + str.Substring(1));
else
    Console.WriteLine("String is too short.");
Console.Write("Enter character to replace: ");
char ch = Convert.ToChar(Console.ReadLine());
string replaced = str.Replace(ch, '$');
Console.WriteLine("Modified String: " + replaced);
string str2 = str;
str2 = str2 + " Welcome";
Console.WriteLine("Original String: " + str);
Console.WriteLine("Second String: " + str2);
}
}

```

OUTPUT

Enter a string: Computer

Reversed String: retupmoC

Substring from 2nd position: omputer

Enter character to replace: o

Modified String: C\$mputer

Original String: Computer

Second String: Computer Welcome